

PT-1.7. Stage 4: Checkpoint 2 and Project Report (Final Demo)

Stage 4: Checkpoint 2 and Project Report (30%)

In the previous stages, we designed all the database components for our application. Now it is time to connect your database to the front-end application. For this stage, your team is expected to build a simple user interface that interacts with the database you developed in Stage 3. Please note that we do not grade based on how fancy your user interface is. We grade based on the functionality of your interface and how it connects to the database.

For this stage, your team is expected to develop advanced database functions and add a creative component to your application. **You are expected to continue working on the same (or an improved version of the) database you developed in Stage 3.** You will also use this time to complete your application interface.

Your web application should be designed for your users, not for the teaching staff. So, make sure that your application functionality matches your intended application description (usefulness, realness, uniqueness,...).

- **Checkpoint 2 (25% of stage 4 grade):** Each team will demo to their project TA by signing up on the demo signup sheet. More details will be posted as the demos approach. All team members are required to be present at the demo. Please note that missing your demo slot will result in a grade of 0 for whoever missed the demo. Other accommodations follow the course syllabus. Please note that the deadline for code submission is before the time of the demos. This is to ensure that all students have the same amount of time to work on the project before the demos. **The code you present during the**

demo must be the same as the code you submitted on Canvas at the time of the deadline for this stage. Any code changes after the deadline will not be graded. Additionally, no changes may be made to the database after the deadline.

- **Project Report (3% of stage 4 grade):** Each team is expected to submit a project reflection report.
- **Project Video (2% of stage 4 grade):** Each team is expected to submit a video showcasing their project.

Here are the detailed requirements for stage 4:

1. Application Requirements:

- Develop a complete application with CRUD (Create, Read, Update, Delete) operations as outlined in your proposal. **This CRUD must be for at least one non-user login table.**
- Implement keyword search functionality, enabling user input and interface results display.

2. Advanced Database Feature:

- Your database should have stored procedure(s), transaction(s), constraints, and trigger(s), accessible via the frontend.

3. Detailed Feature Requirements:

- **Transaction:** Functional, with correct isolation level, at least two advanced queries, and application utility. Advanced queries involve **at least two** of the following SQL concepts:
 - Join multiple relations
 - SET Operators
 - Aggregation via GROUP BY
 - Subqueries that cannot be easily replaced by a join.
- **Stored Procedure:** Functional, with at least two advanced queries, cursors (optional), control structures, and application utility.
- **Trigger:** Functioning triggers, involving an event, condition (IF statement), and action (Update, Insert, Delete), enhancing the application.

- **Constraints:** This requirement can be simply satisfied by defining the appropriate primary keys, and foreign keys. But we strongly encourage the team to use other types of constraints, such as attribute-level, tuple-level, and assertions.
- Implement features in SQL (not with Object Relational Mapping, ORM. More details in the next section), ensuring relevance to the application.

4. **Extra Credit Options:**

- Host **both** application and database on GCP (1%). Doing one or the other will not result in extra credit.
- Develop a creative, relevant function, like interactive visualization or machine learning recommendation (up to 2%). This creative feature should entail the development of complex and sophisticated code, going beyond merely utilizing existing libraries.

5. **Demonstration and Discussion (Checkpoint 2):**

- Conduct a thorough demo, covering the application's purpose, user interaction, design choices, and improvement plans.
- Discuss implementation challenges, design deviations, and potential NoSQL integration.

6. **Project Report:**

- Please list out changes in the directions of your project if the final project is different from your original proposal (based on your stage 1 proposal submission).
- Discuss what you think your application achieved or failed to achieve regarding its usefulness.
- Discuss if you changed the schema or source of the data for your application
- Discuss what you change to your ER diagram and/or your table implementations. What are some differences between the original design and the final design? Why? What do you think is a more suitable design?
- Discuss what functionalities you added or removed. Why?
- Explain how you think your advanced database programs complement your application.

- Each team member should describe one technical challenge that the team encountered. This should be sufficiently detailed such that another future team could use this as helpful advice if they were to start a similar project or where to maintain your project.
- Are there other things that changed comparing the final application with the original proposal?
- Describe future work that you think, other than the interface, that the application can improve on
- Describe the final division of labor and how well you managed teamwork.

7. Project Video:

- Please create a roughly 15 minute video showcasing your application. Treat it as a pitch you would show to potential users or investors. We would like to use these videos to show off cool projects to future students, so put your best foot forward!
- The video should **not** include the code or infrastructure. Users and investors wouldn't care about that!

8. Grading Rubric:

- **Checkpoint 2:**
 - **Presentation:** Timeliness and completion within scheduled time (+1%).
 - **CRUD Operations:** The application functionality includes reads, insertions, updates, and deletions from the database (1.5% each, 6% total).
 - **Keyword Search:** Interface, code, and query execution (5% total).
 - **Advanced Program:** Transaction(s), trigger(s), constraints, and stored procedure(s) are implemented and integrated into the application (4% for the transaction, 4% for the trigger, 4% for the stored procedure, 1% for the constraint, 13% total), with detailed criteria for transactions, stored procedures, and triggers. Additionally, teams must include a file in their docs folder that contains the code for their transaction, trigger, and stored procedure.
 - **Extra Credit:** application and database hosted on GCP and creative component relevance (up to +3% total).

- **Final Report:** Your project reflection report should cover all required topics. The report will be graded on completeness and correctness. (3%).
- **Final Video:** Your project video should be roughly 15 minutes long and cover all the required components. (2%).

9. Additional Requirements:

- Discuss advanced database program choice and creative element's value at the demo.
- Submit code to the repository, and tag release submitted to Canvas (failure to do so results in a 3% deduction).
- Full team presence at the demo (absence results in a zero for that student).

10. Penalties:

- Missing points or incorrect responses during the demo (-1% per point).
- Failure to submit any code to the repo by the deadline results in a zero for this stage.
- Trying to demo code that is not from your repo at the time of the deadline results in a zero for this stage.
- Trying to demo when changes have been made to the database after the deadline results in a penalty (-5%).
- Use of ORMs results in a zero for this stage.

In addition, if you lost points on Stage 3, you will be allowed to revise your Stage 3 to get points back for this stage. To do so, you will need to do the following:

1. Create new versions of the required documents for Stage 3 that contain the necessary revisions
2. Create a new file named `stage3_revisions.md` that explains what parts of the original submission were changed and which comments they address
3. Include the documents in your stage.4 release

These changes will allow you to earn up to 80% of the points you lost from Stage 3. For instance, if you got 25/30 on Stage 3, losing 5 points, if you address all comments you can earn up to 29/30, or 4 points back. **Failure to follow these instructions will make you ineligible for earning points back.** These points can only be earned back if you make revisions by the Stage 4 deadline.

Save

PT -1

Assessment overview

Total points: 0/7

Score:

0%

Question PT-1.7

Total points: — /1

Manually-graded question

Report an error in this question 

Previous question

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 Personal Notes

No attached notes

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